Linear Algebra — Mid-Term Model Test 1

Covers Sections 1.1–1.3 and 2.1–2.6 · Level: standard (same as the sample test)

Instructions. Time allowed: 120 minutes. Calculators are *not* allowed. Show all working. Total: 100 points. Difficulty: standard (same level as the sample test).

1 Vectors

Let $v = (1, 2, 2)$ and $w = (3, -1, 1)$.

(a) Compute $2v - 3w$ and $v \cdot w$. **[5 pts]**

(b) Find $\|v\|$ and the unit vector u in the direction of v. **[5 pts]**

(c) Find the angle between $a = (1, 1, 0)$ and $b = (1, 0, 1)$. **[5 pts]**

(d) Find the number c such that $(1, c, 2)$ is perpendicular to w. **[5 pts]**

2 Matrix operations

(1) Calculate the multiplication **[12 pts]**

$$\begin{pmatrix} 2 & -1 & 1 & 0 \\ 1 & 3 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -2 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ -3 & 1 \end{pmatrix}.$$

(2) Calculate $f(A)$, where **[12 pts]**

$$A = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}, \qquad f(x) = x^3 - x^2 + 2x + 1.$$

3 Elimination

Solve the system by elimination. State the multipliers ℓ_{ij} you use and the pivots. **[18 pts]**

$$\begin{cases} 2x + y + z = 5, \\ 4x - 6y = -2, \\ -2x + 7y + 2z = 9. \end{cases}$$

4 Inverse matrix

Calculate the inverse of the following matrix by Gauss–Jordan elimination. **[18 pts]**

$$\begin{pmatrix} 1 & 0 & -1 \\ -1 & -1 & 1 \\ 1 & -2 & -2 \end{pmatrix}.$$

5 $A = LU$ factorization

Factor the following matrix as $A = LU$, where L is lower triangular with 1's on the diagonal and U is upper triangular. **[20 pts]**

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 4 \\ 3 & 8 & 9 \end{pmatrix}.$$